

Computer Science & IT

Discrete Mathematics



Graph Theory

Lecture No. 10



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Recap of Previous Lecture



Topic

Chromatic number



Topic

Welsh Powell's algorithm for vertex coloring



Topic

Matching



Topics to be Covered





Topic : Maximal Matching

- A matching of graph G is called a maximal matching, if no other edge of graph G can be added to that matching without destroying its property of being a matching.

In above example,

M_1 , M_2 and M_3 are maximal matching of graph G .



Topic : Maximum Matching

Largest Maximal Matching

- The matchings of graph G with maximum number of edges of graph G , are called maximum matching.
There may be more than one maximum matching for the same graph.

In the above example,

M_1 and M_2 are maximum matching of graph G .

$\{ M_3 \text{ is a maximal matching} \}$
 $\{ \text{but } M_3 \text{ is not a maximum matching of graph } G \}$

Note:- Every maximum matching of graph G
is also a maximal matching of that graph,
but Every maximal matching of graph G
need not be a maximum matching {eg: M_3 }



Topic : Matching Number

★ Matching Number of graph G = Number of edges in one of the maximum matching of graph G .

In the above example,

$$\text{Matching no. of } G = \text{No. of Edges in } M_1 \text{ or } M_2 = 2$$

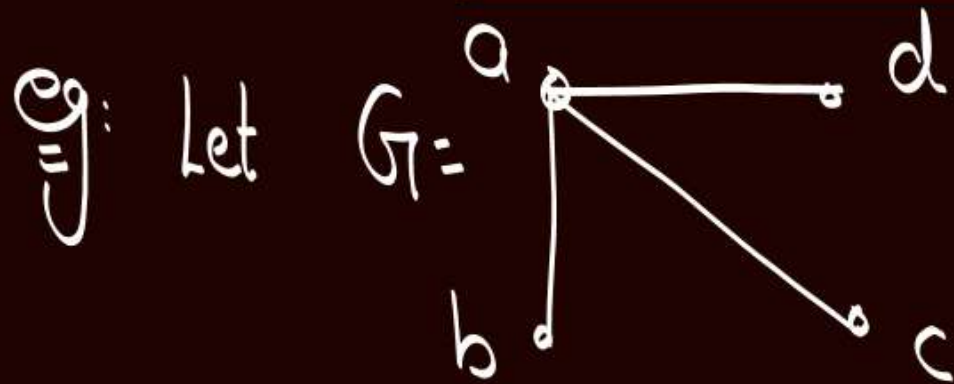
because M_1 & M_2 are maximum matching

Note:-

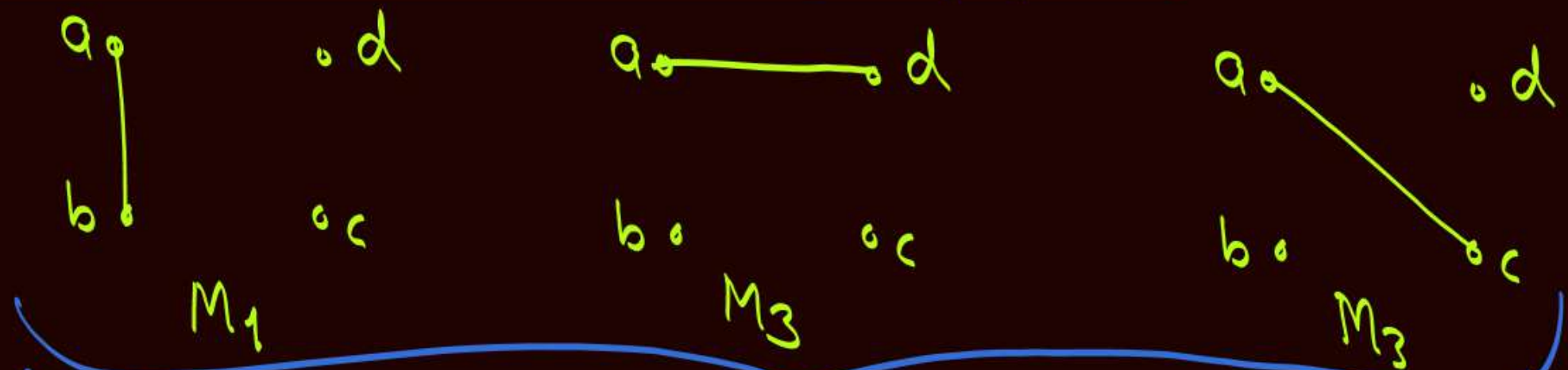
Let G be a graph with n -vertices,
then

$$\text{matching number of graph } G \leq \left\lfloor \frac{n}{2} \right\rfloor$$

It is actually
An upper bound
On matching No.
of a graph.



$$n = 4$$
$$\left\lfloor \frac{n}{2} \right\rfloor = \left\lfloor \frac{4}{2} \right\rfloor = 2$$



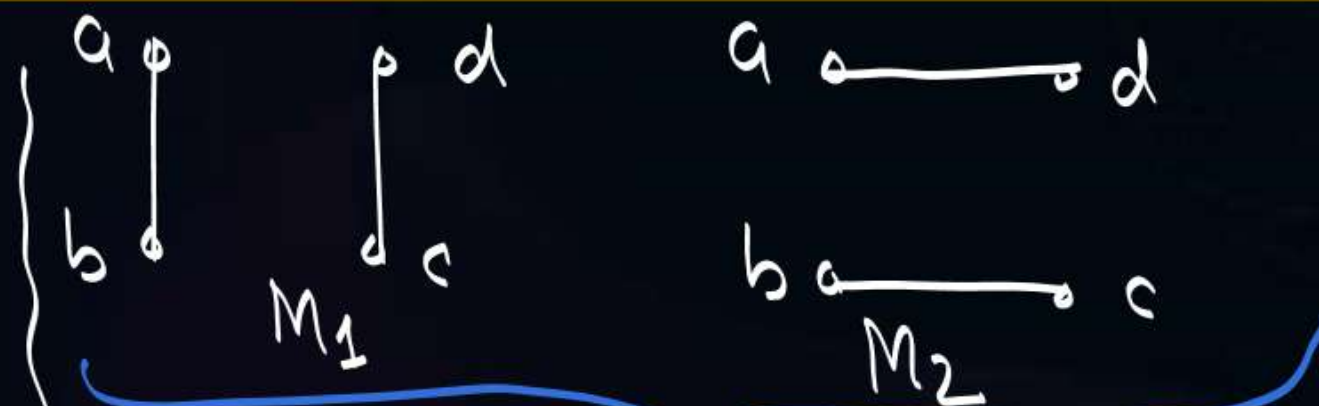
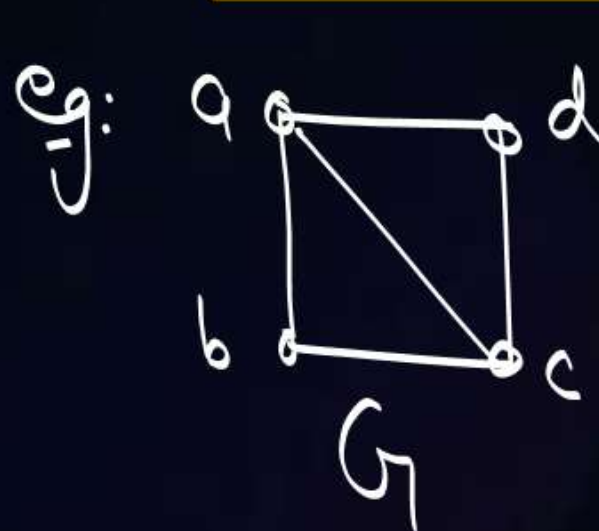
All three are maximal as well as maximum matching of G
∴ Matching no. of $G = 2 \leq \left\lfloor \frac{n}{2} \right\rfloor$



Topic : Perfect Matching



A matching of graph G is called a perfect matching if and only if degree of each vertex in the matching of graph G is '1'.
i.e. When all vertices in a matching are "matching vertex".
in a perfect matching, $\deg(v) = 1, \forall v \in G$

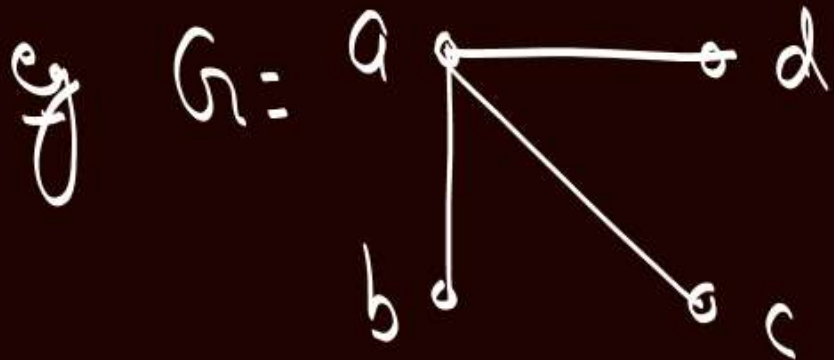


Both M_1 & M_2 are perfect matching of graph G .

Note:-

① Every perfect matching of graph G , is also a maximum matching of graph G .

② But, Every maximum matching of graph G need not be a perfect matching

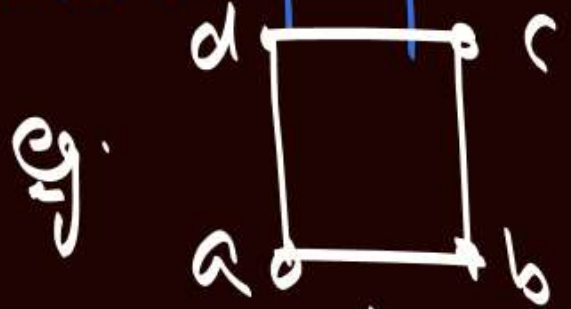


} No maximum matching of graph G is a perfect matching

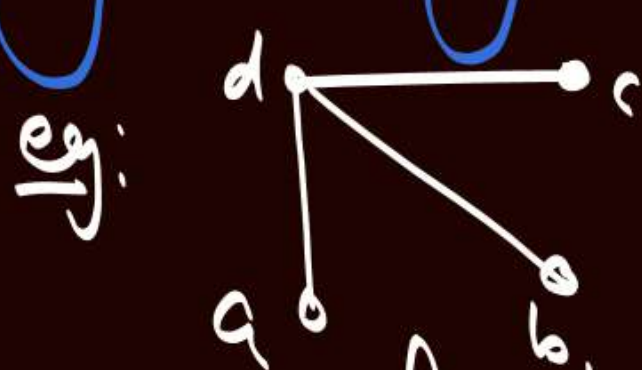
Note:- ① Perfect matching need not exist for every graph

② In a graph G , if number of vertices in G are 'odd', then Perfect matching never exist for graph G

③ If number of vertices in graph G are even, then perfect matching may or may not exist for graph G



No. of vertices = 4
And perfect matching exist



No. of vertices = 4 = even,
but No perfect matching

Note:-

If perfect matching exists for graph G ,
then number of vertices in graph G are even.

✓ (Converse of the statement need not be true)

Note:-

Let G be a graph with n -vertices,

If perfect matching exists for graph G , then $n = \text{even}$.

And Matching Number of graph $G = \frac{n}{2}$

★ For a Complete graph K_n perfect matching exists if and only if $n = \text{Even}$ in K_n

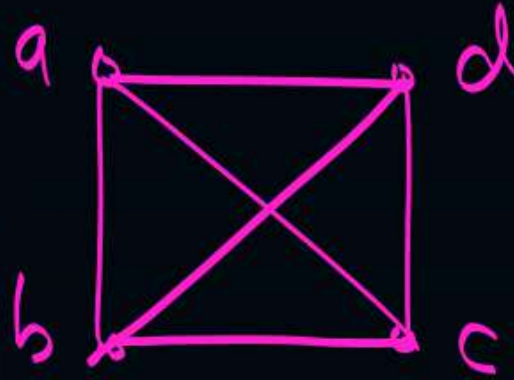
★ In a Complete bipartite graph $K_{m,n}$ perfect matching exist if and only if $m = n$



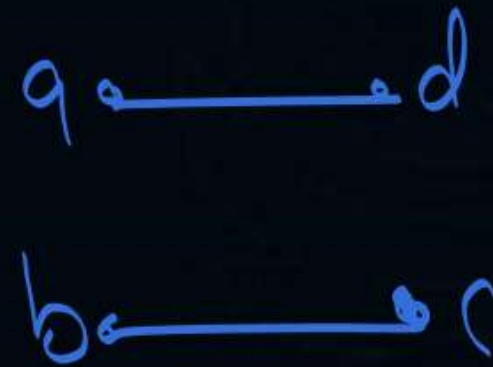
Topic : Perfect Matching

Find the number of perfect matching in a complete graph K_n (n is even)

Note: for graph $K_4 =$

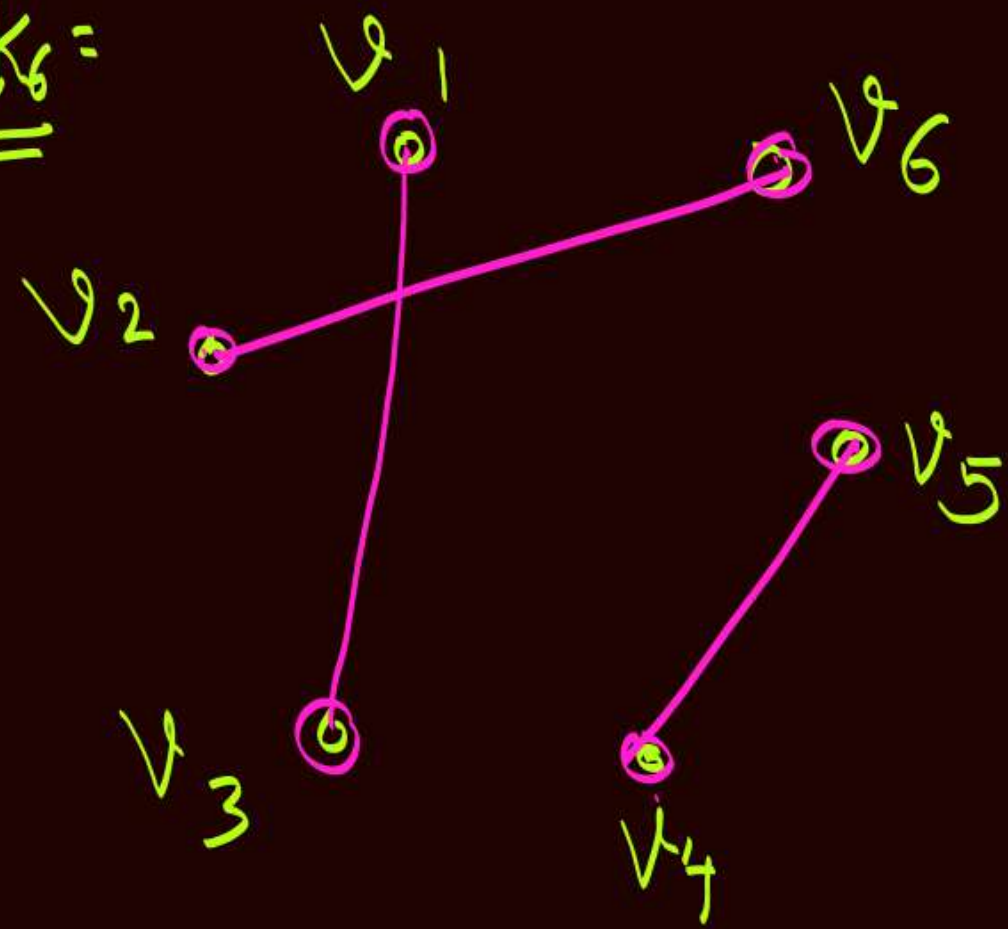


&



are different
Perfect matching

Let $K_6 =$



5 ways * 3 ways * 1 way

$$\begin{aligned} \therefore \# \text{ Perfect Matching of } K_6 &= \overset{(6-1)}{\uparrow} 5 * \overset{(6-3)}{\uparrow} 3 * \overset{(6-5)}{\uparrow} 1 \dots \\ &= \boxed{15} \end{aligned}$$

Number of perfect matching of K_n { $n = \text{Even}$ }

$$\text{Number of perfect matching of } K_n = (n-1) \cdot (n-3) \cdot (n-5) \dots 5 \cdot 3 \cdot 1$$

$$= (n-1) \cdot (n-3) \cdot (n-5) \dots 5 \cdot 3 \cdot 1 * \frac{n \cdot (n-2) \cdot (n-4) \dots 6 \cdot 4 \cdot 2}{n \cdot (n-2) \cdot (n-4) \dots 6 \cdot 4 \cdot 2}$$

$$= \frac{n \cdot (n-1) \cdot (n-2) \cdot (n-3) \dots 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{n \cdot (n-2) \cdot (n-4) \dots 6 \cdot 4 \cdot 2}$$

$$= \frac{n!}{(2 * \frac{n}{2}) (2 * \frac{n-2}{2}) (2 * \frac{n-4}{2}) \dots (2 * \frac{6}{2}) (2 * \frac{4}{2}) (2 * \frac{2}{2})}$$

$$= \frac{n!}{(2 * (\frac{n}{2})) (2 * (\frac{n}{2} - 1)) (2 * (\frac{n}{2} - 2)) \dots (2 * 3) (2 * 2) (2 * 1)}$$

$$= \frac{n!}{(2)^{n/2} \cdot (\frac{n}{2}) \cdot (\frac{n}{2} - 1) \cdot (\frac{n}{2} - 2) \dots 3 \cdot 2 \cdot 1} = \frac{n!}{(2)^{n/2} \cdot (\frac{n}{2})!}$$

* Number of Perfect matching of Complete graph K_{2n}
just replace n by $(2n)$
 $\underbrace{K_{2n}}_{\substack{\text{always even} \\ \text{for all } n \in \mathbb{N}}}$

$$\therefore \# \text{ perfect matching} = \frac{(2n)!}{(2)^{\left(\frac{2n}{2}\right)} \cdot \left(\frac{2n}{2}\right)!}$$

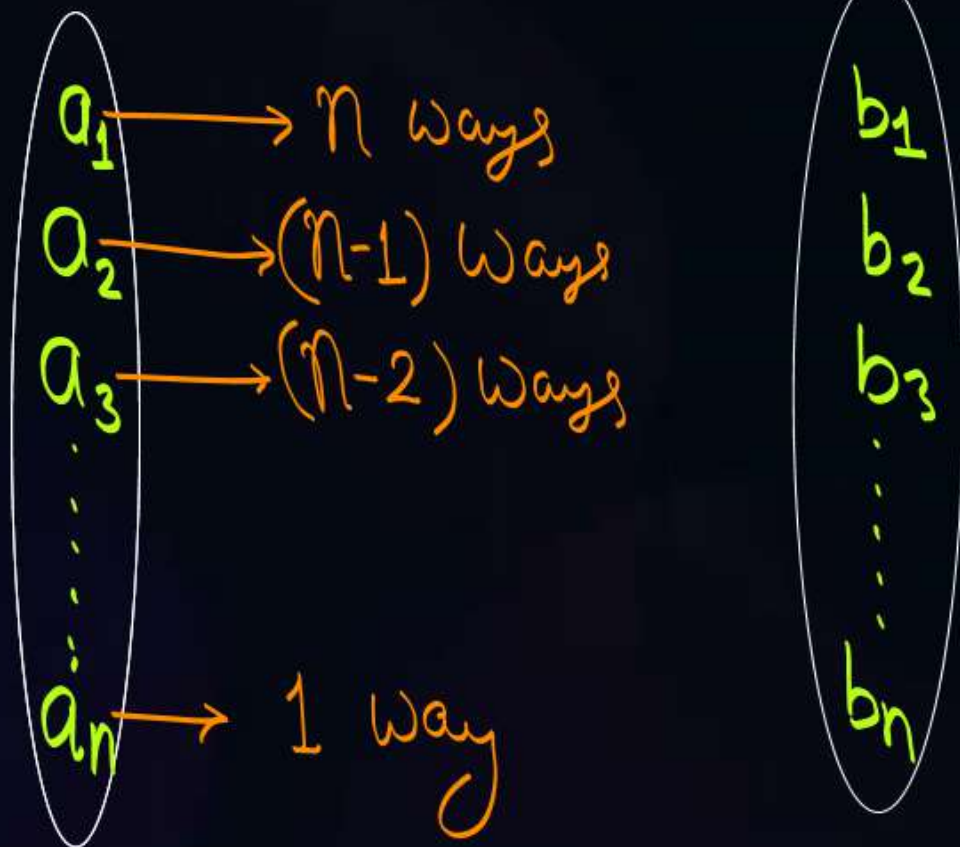
$$= \frac{(2n)!}{(2)^n \cdot n!}$$



Topic : Perfect Matching

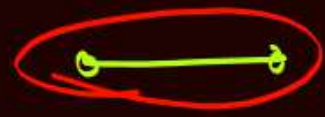
Find the number of perfect matching in a complete bipartite graph $K_{n,n}$

$V_1, |V_1|=n$ $V_2, |V_2|=n$



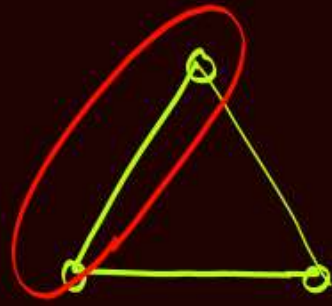
$$\therefore \# \text{ Perfect Matching of } K_{n,n} = n * (n-1) * (n-2) \cdot \cdot \cdot 3 \cdot 2 \cdot 1$$
$$= n!$$

Q:- Find the matching number of Complete graph K_n



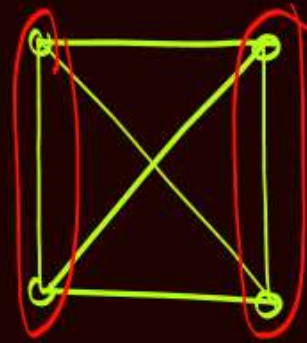
K_2

Matching No.
of $K_2 = 1$



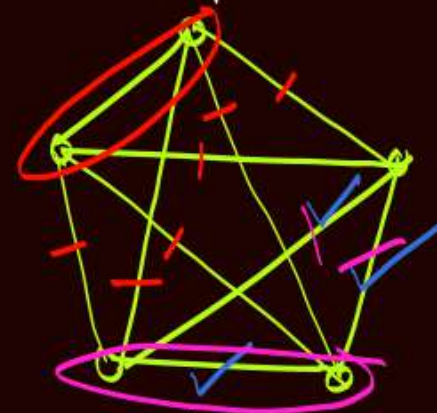
K_3

Matching No.
of $K_3 = 1$



K_4

Matching No.
of $K_4 = 2$



K_5

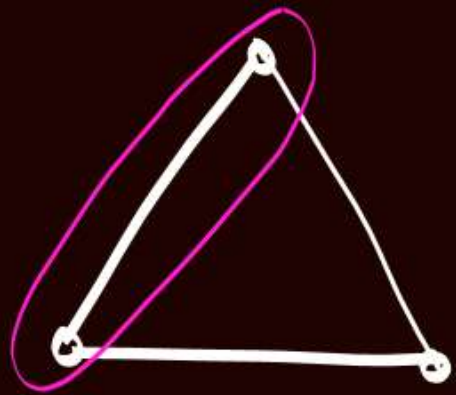
Matching No.
of $K_5 = 2$

$$\text{Matching No. of } K_n = \begin{cases} \frac{n}{2} & \text{if } n = \text{Even} \\ \frac{(n-1)}{2} & \text{if } n = \text{odd} \end{cases}$$

i.e.

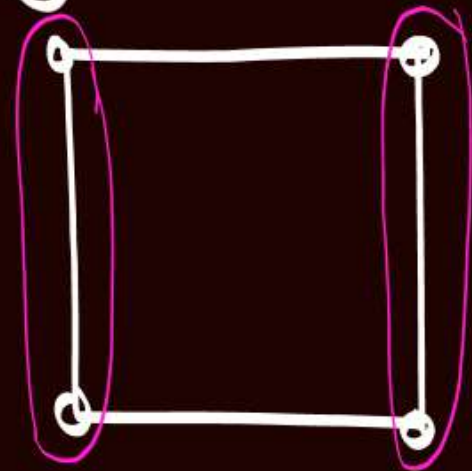
$$\text{Matching No. of } K_n = \left\lfloor \frac{n}{2} \right\rfloor$$

Q: Find matching number of Cycle graph C_n



C_3

Matching No. = 1



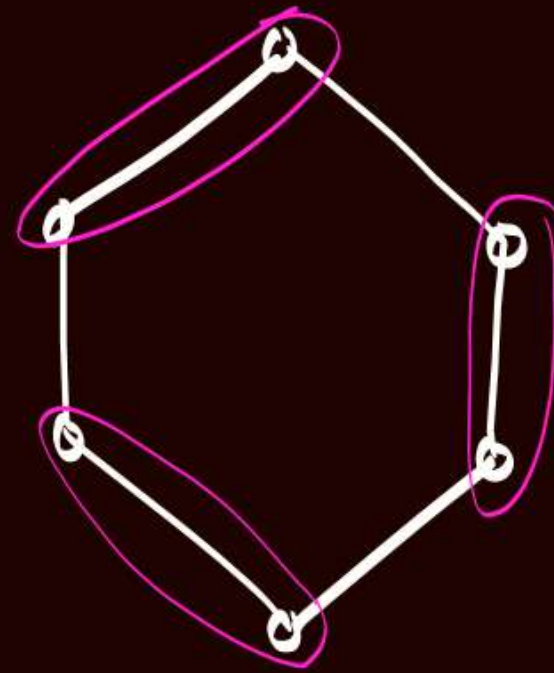
C_4

Matching No. = 2



C_5

Matching No. = 2

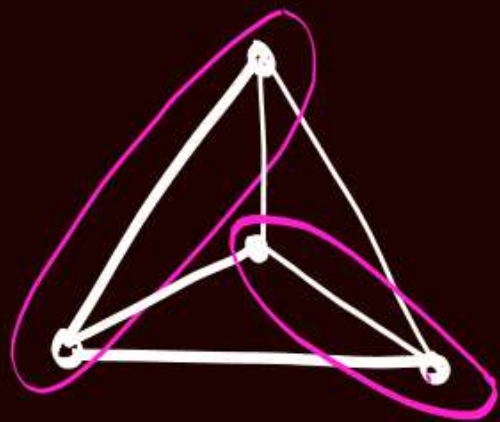


C_6

Matching No. = 3

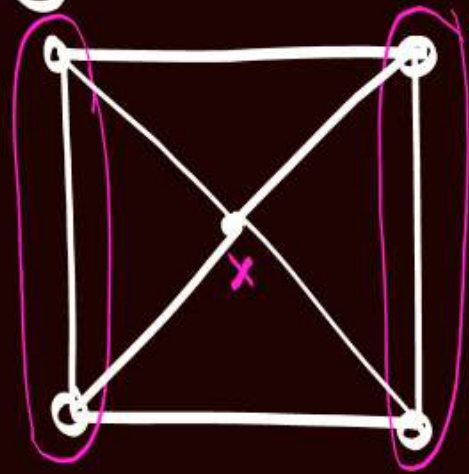
$$\text{Matching no. of } C_n = \left\lfloor \frac{n}{2} \right\rfloor$$

Q: Find matching number of wheel graph W_n



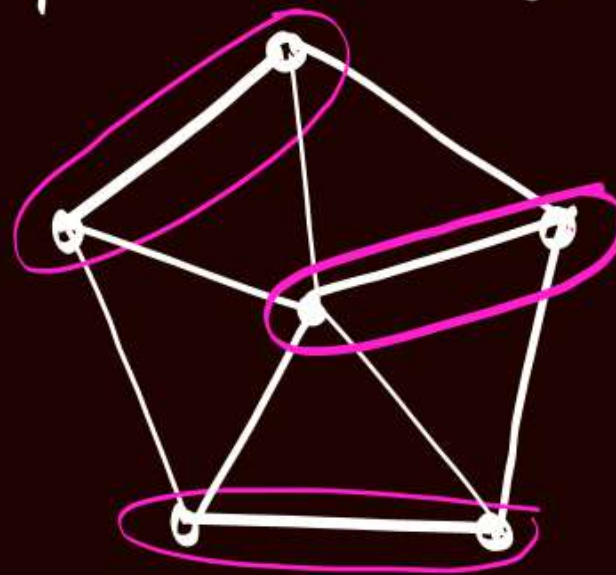
$$W_4 = C_3 + 1$$

Matching No. = 2



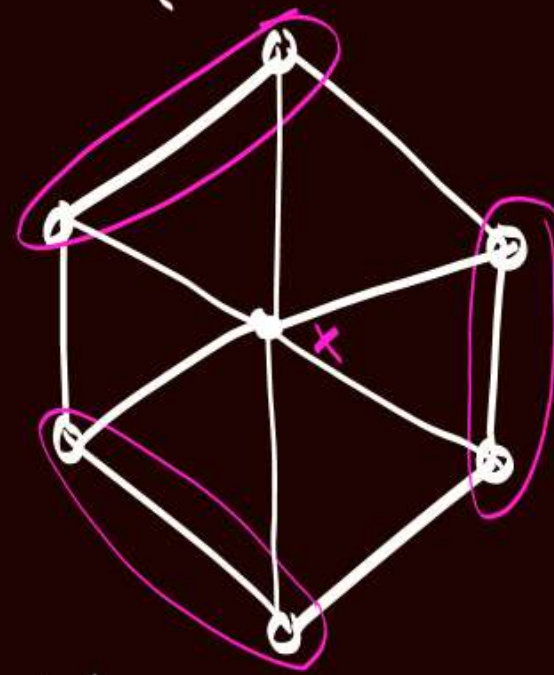
$$W_5 = C_4 + 1$$

Matching No. = 2



$$W_6 = C_5 + 1$$

Matching No. = 3



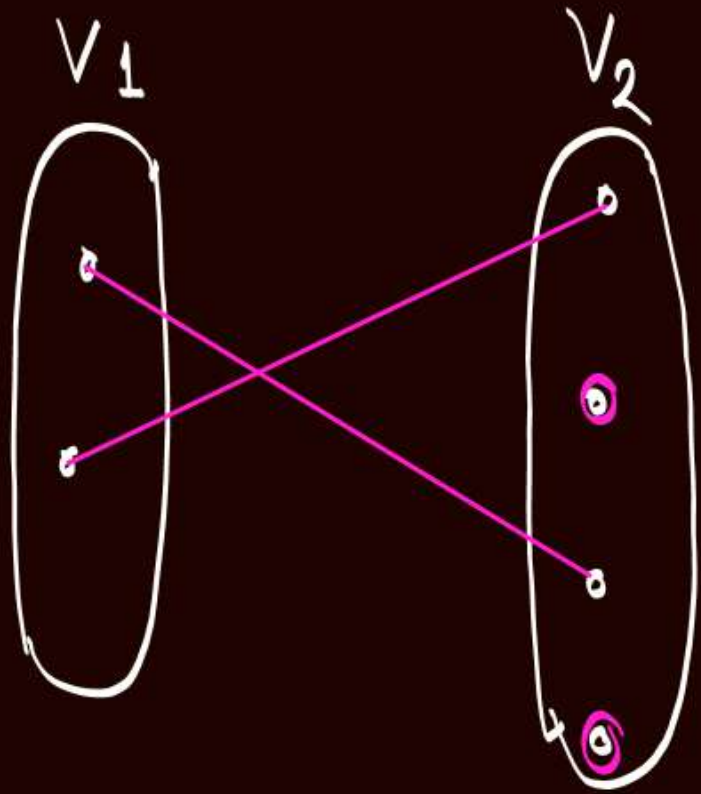
$$W_7 = C_6 + 1$$

Matching No. = 3

$$\text{Matching no. of } W_n = \left\lfloor \frac{n}{2} \right\rfloor$$

Q:- Find the matching number of Complete bipartite graph $K_{m,n}$

Soln



Matching number of
Complete bipartite graph
 $K_{m,n} = \text{Min}(m,n)$

Matching No. of $K_{2,4} = 2$
 $\uparrow$
 $\text{Min}(2,4)$

Q. Find the matching number of star graph with 'n' vertices

• Star graph with n-vertices $\equiv K_{1, n-1} \text{ or } K_{n-1, 1}$

$\therefore$ Matching No. of
Star graph with n-vertices $= \min(1, n-1)$
 $= 1$ Ans



2 mins Summary



✓
Topic

Maximal matching

✓
Topic

Maximum matching

✓
Topic

Matching number

✓
Topic

Perfect matching

THANK - YOU